

CAPE LISBURNE, AK, USA

WMO: 701040

Lat: 68.875N

Lon: 166.113W

Elev: 5

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26631

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.8	-37.2	0.1	-32.8	-34.9	0.1	-30.7	24.1	-6.5	20.8	-6.6	3.3	110	1.040

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	16.4	12.3	14.5	11.4	13.0	10.7	13.0	15.2	11.8	14.0	10.8	12.9	6.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	8.6	14.1	10.5	7.9	12.9	9.5	7.4	12.0	36.5	15.3	33.6	14.1	31.3	13.0	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
18.4	16.0	14.2	DB	-33.1	20.0	4.9	2.3	-36.6	21.7	-39.5	23.0	-42.2	24.4	-45.7	26.1	26.1
			WB	-33.2	14.6	4.3	1.2	-36.3	15.5	-38.9	16.2	-41.3	16.9	-44.5	17.8	17.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-18.7	-18.0	-18.7	-11.0	-1.4	4.0	9.0	8.6	4.5	-1.7	-9.0	-15.3
	DBStd	11.88	8.23	9.10	7.84	6.21	4.32	3.62	2.99	2.81	3.44	4.09	6.08	7.39
	HDD10.0	5732	891	784	891	629	353	186	54	62	167	363	569	784
	HDD18.3	8724	1149	1018	1149	879	612	431	288	302	414	621	819	1043
	CDD10.0	49	0	0	0	0	0	5	24	18	3	0	0	0
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.8	5.8	6.1	5.6	5.3	4.8	4.3	5.6	5.8	6.4	6.9	6.7	6.2
Precipitation	PrecAvg	462	30	31	19	19	18	28	57	79	53	58	41	29
	PrecMax	627	76	77	57	53	44	50	98	149	90	98	116	58
	PrecMin	314	3	3	3	3	0	12	18	18	11	19	7	3
	PrecStd	78	18	20	15	14	11	9	19	32	19	20	26	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	4.0	2.1	5.7	10.6	18.2	18.4	21.3	14.5	9.7	6.1	4.9
		MCWB	1.2	1.6	-0.9	2.3	6.8	11.8	13.4	18.0	10.6	5.4	4.1	0.8
	2%	DB	1.0	1.1	0.1	3.2	7.3	14.1	16.5	15.7	12.1	6.9	2.8	1.2
		MCWB	-0.3	-0.2	-1.9	1.4	4.8	10.4	12.4	13.1	9.7	4.3	1.5	-0.6
	5%	DB	-1.6	-0.6	-1.8	1.0	5.5	11.5	14.9	13.6	10.8	4.9	1.1	-1.0
		MCWB	-2.9	-1.7	-3.5	-0.4	3.8	8.5	11.6	11.7	8.9	3.5	0.1	-2.2
	10%	DB	-4.9	-3.2	-5.1	-2.0	3.8	9.4	13.6	12.4	9.7	3.3	-0.9	-4.0
		MCWB	-5.9	-4.4	-6.4	-3.0	2.5	7.2	10.9	10.8	8.1	2.1	-1.9	-5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.5	-0.6	2.8	6.9	12.5	14.2	18.5	11.4	6.1	4.0	1.7
		MCDB	2.8	3.4	0.4	4.9	9.9	16.9	16.7	21.0	13.2	8.5	5.8	4.1
	2%	WB	-0.4	-0.1	-2.2	1.3	5.2	10.6	12.9	13.5	10.2	4.7	1.6	-0.4
		MCDB	0.8	1.0	-0.8	3.2	6.9	14.3	15.5	15.3	11.9	6.4	2.5	1.5
	5%	WB	-2.8	-1.7	-4.1	-0.3	3.7	8.8	11.9	12.0	9.2	3.5	0.2	-2.2
		MCDB	-1.8	-0.7	-2.4	1.0	5.2	11.3	14.4	13.5	10.7	4.8	1.1	-0.9
	10%	WB	-6.1	-4.5	-7.3	-2.9	2.5	7.3	11.0	10.8	8.0	2.2	-1.9	-5.0
		MCDB	-5.0	-3.3	-6.0	-2.0	3.7	9.5	13.3	12.3	9.5	3.5	-1.0	-4.0
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.3	5.5	5.6	4.1	4.8	4.7	3.5	3.1	3.2	4.2	4.8
		MCDBR	8.7	7.9	6.7	7.3	7.0	8.4	7.1	5.5	4.6	4.4	4.6	6.8
		MCWBR	7.9	7.1	6.2	6.1	5.0	5.4	4.7	3.9	3.3	3.6	3.8	5.8
	5% WB	MCDBR	8.6	8.0	7.5	7.3	6.6	8.0	6.5	5.2	4.4	4.2	4.4	7.0
		MCWBR	7.8	7.3	6.6	6.3	4.8	5.4	4.6	3.9	3.2	3.5	3.8	6.1
Clear-Sky Solar Irradiance	taub	0.162	0.221	0.252	0.323	0.356	0.343	0.359	0.367	0.310	0.253	0.358	N/A	
	taud	1.903	2.047	2.188	2.010	2.061	2.399	2.452	2.389	2.498	2.212	2.848	N/A	
	Ebn at Noon	75	556	786	799	817	854	825	767	732	510	123	0	
	Edh at Noon	5	44	74	123	132	98	91	86	59	37	11	0	
All-Sky Solar Radiation	RadAvg	0.02	0.45	2.13	4.31	5.37	5.81	4.60	3.03	1.72	0.52	0.06	0.00	
	RadStd	0.00	0.05	0.16	0.25	0.64	0.46	0.46	0.38	0.17	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page